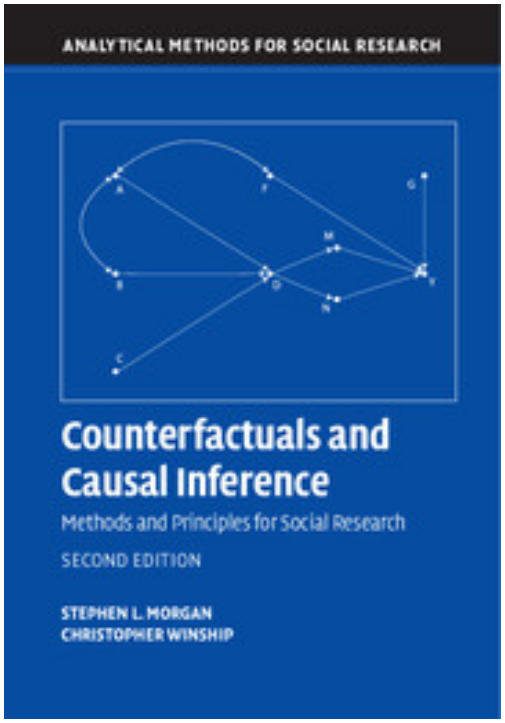
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 3

Outline

1. Identification
2. Basic Elements of Causal Graphs
3. Graphs and Structural Equations
4. Causal Graphs and the Potential Outcome Model
5. Appendix to Chapter: Graphs, Interventions, and Potential Outcomes

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Two inputs into an identification analysis

1. The set of assumptions about causal relationships that the analyst is willing to assert based on theory and past research, including assumptions about relationships between variables that have not been observed but that are related to both the cause and the outcome of interest.
2. The pattern of information that one can assume would be contained in the joint distribution of the variables in the observed dataset *if* all members of the population had been included in the sample that generated the dataset.

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Nodes, edges, paths, and cycles

- Each *node* of a graph represents a random variable and is labeled by a letter, such as A, B, or C. Nodes that are represented by a solid circle $\bullet$ are observed random variables, whereas nodes that are represented by a hollow circle $\circ$ are unobserved random variables.
- Causal effects are represented by directed edges $\rightarrow$ (i.e., single-headed arrows).
- A *path* is any sequence of edges pointing in any direction that connects one variable to another.
- A *directed path* is a path in which all edges point in the same direction.
- A variable is a *descendant* of another variable if it can be reached by a directed path. for $A \rightarrow B$, the variable A is the *parent* while the variable B is the *child*.

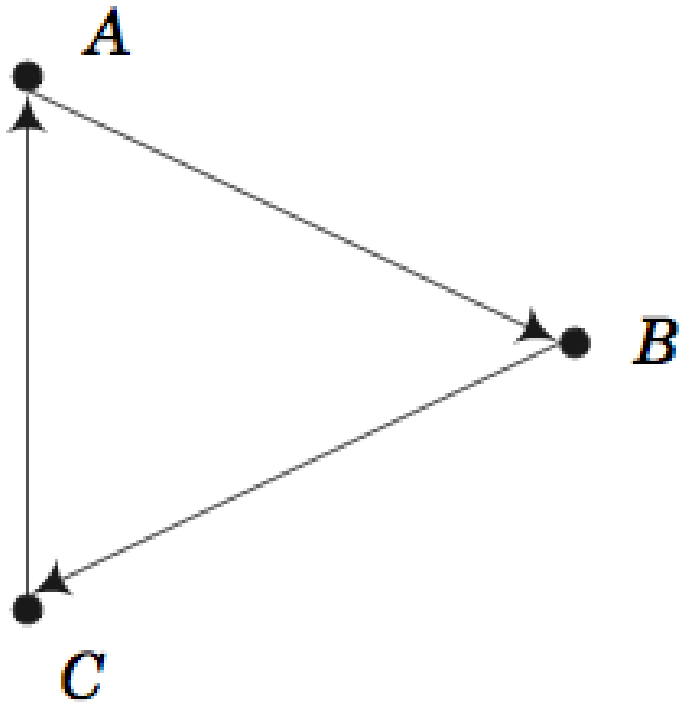


Figure 3.1 A directed graph that includes a cycle.

In this book, we will consider only a subset of directed graphs known as *directed acyclic graphs* or DAGs.

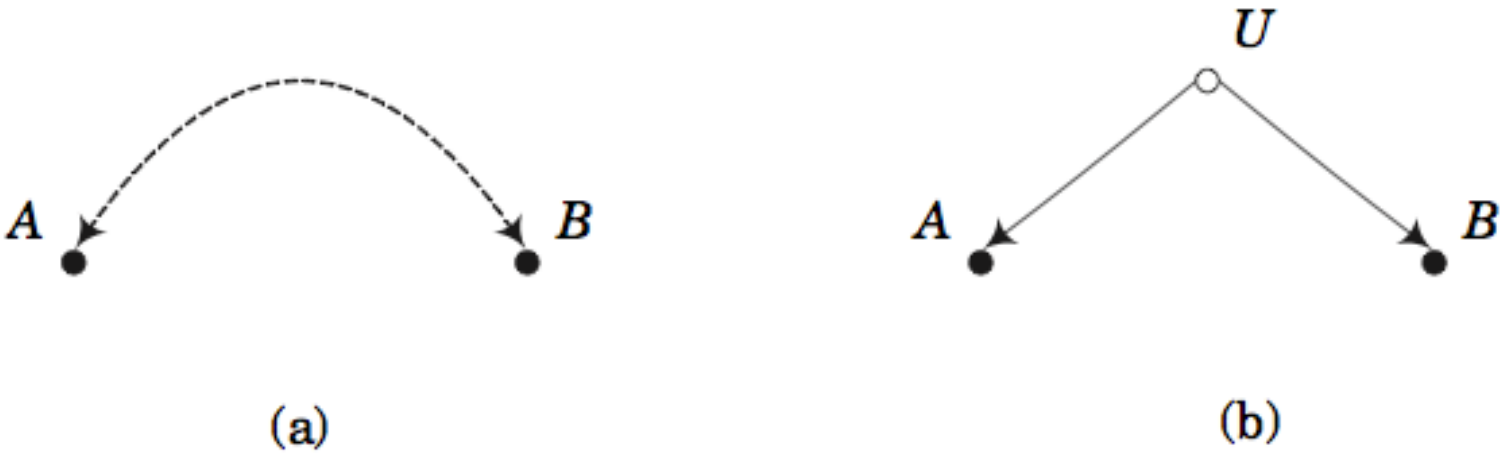
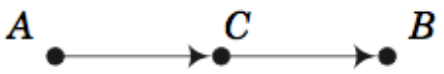
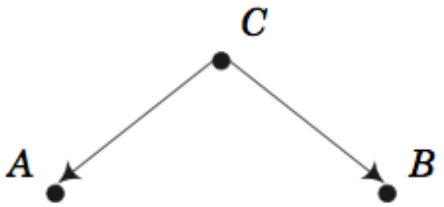


Figure 3.2 Two representations of the joint dependence of A and B on unobserved common causes.

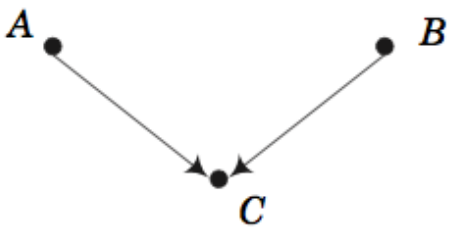
Causal graphs for three variables



(a) Mediation



(b) Mutual dependence



(c) Mutual causation

Figure 3.3 Basic patterns of causal relationships for three variables.

Colliders

- A variable is a *collider* along a particular path if it has two arrows pointing directly at it.
- Collider variables are common in social science applications: Any endogenous variable that has two or more causes is a collider along some path.
- A path that is connected by a collider variable does not generate an unconditional association between the variables that cause the collider variable.
- But ... we will learn to pay very close attention to colliders nonetheless ...

A first look at confounding and conditioning

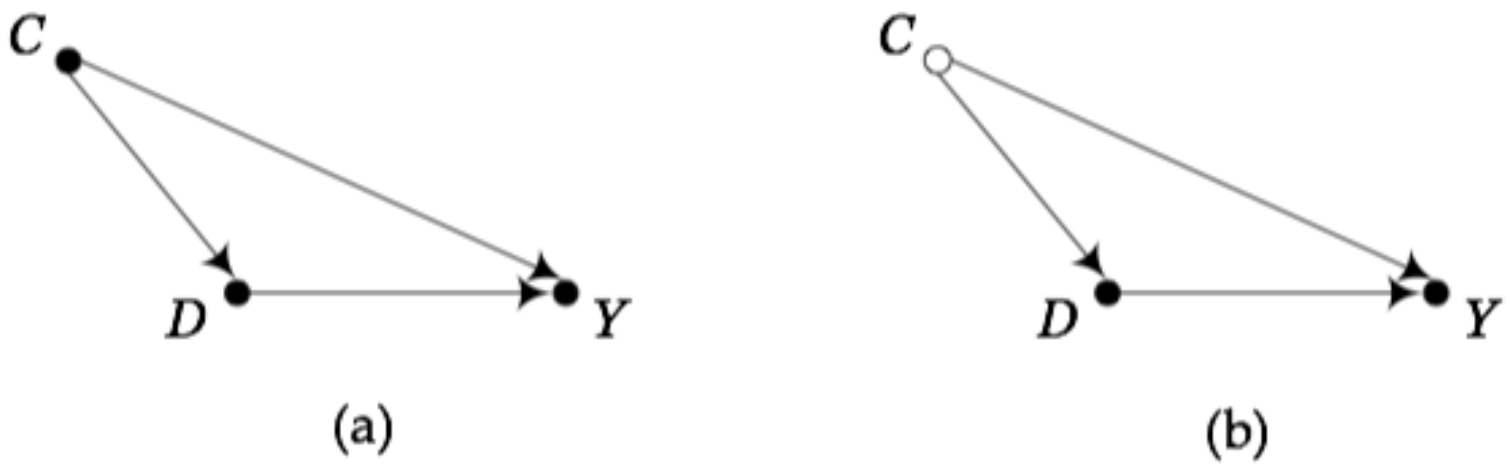


Figure 3.4 Two graphs in which the causal effect of D on Y is confounded by C.

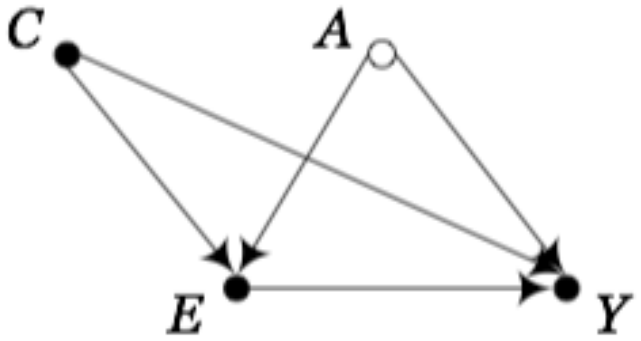


Figure 3.5 A causal graph in which the effect of education (E) on earnings (Y) is confounded by observed variables (C) and by unobserved ability (A).

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The appeal and critique of path diagrams

- Graphs have been used for decades in social science research
- The new causal graph methodology is connected to this tradition
- It is useful to review what would have been taught in a course such as this one in the 1970s

A traditional linear additive path diagram

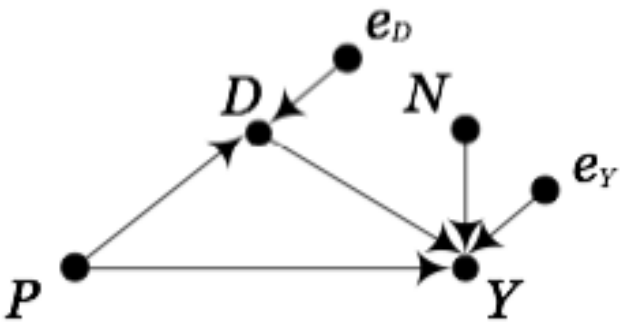


Figure 3.6 A traditional linear additive path diagram for the effects of parental background (P), charter schools (D), and neighborhoods (N) on test scores (Y).

Structural equations for the two endogenous variables

Path model equations:

$$D = a_D + b_P P + e_D, \tag{3.1}$$
$$Y = a_Y + b_D D + b_P P + b_N N + e_Y. \tag{3.2}$$

These structural equations are linear and additive in the variables P, D, and N, and each equation has terms, e_D and e_Y, that are represented in the path diagram as all other determinants of D and Y other than P, D, and N.

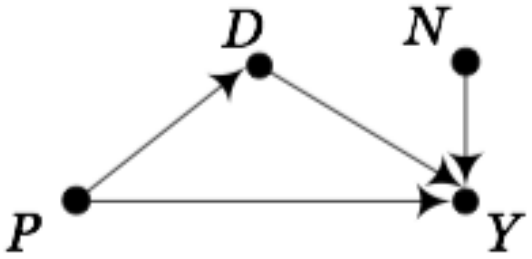
An elaborated regression specification

In response to a puzzled student:

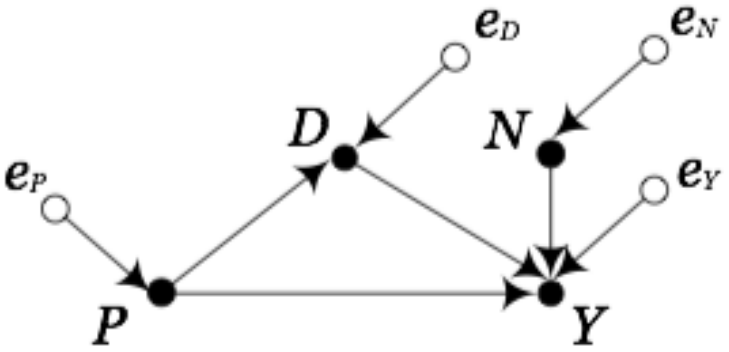
$$Y = a_Y + b_D D + b_P P + b_{D\times P}(D \times P) + b_N N + b_{D\times N}(D \times N) + e_Y. \tag{3.3}$$

In this case, the path diagram ceases to represent an underlying set of structural causal relationships and is instead best interpreted as only a simplified reflection of a more specific regression model.

"New" causal graphs and magnification



(a) the standard representation



(b) under magnification

Figure 3.7 Equivalent directed graph representations of the effects of parental background (P), charter schools (D), and neighborhoods (N) on test scores (Y).

The shift to nonparametric structural equations

Nonparametric structural equations (i.e., no functional form):

$$P = f_P(e_P), \tag{3.4}$$
$$D = f_D(P, e_D), \tag{3.5}$$
$$N = f_N(e_N), \tag{3.6}$$
$$Y = f_Y(P, D, N, e_Y). \tag{3.7}$$

What are these terms, e_P, e_D, e_N, and e_Y? They represent all causes of P, D, N, and Y, respectively, that can be regarded as "idiosyncratic" causes of each variable. They are assumed to be independent of each other and of P, D, N, and Y. As such, they can be suppressed in the standard representation of a directed graph.

How should we think about these "error" terms?

Very carefully:

- The researcher must be able to assume that the structural error terms are mutually independent of each other and of all of the other variables in the graph, such that no pair is mutually dependent on a common third variable that has been mistakenly omitted from the graph.
- If this assumption is dubious, then the researcher must re-draw the graph including hollow nodes for any mistakenly omitted unobserved common causes.
- These unobserved variables must then be given directed edges that point to the variables that they cause.
- The new error terms for the re-drawn graph can then be redefined so that they can be pushed into the background (but still rendered visible under magnification).

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The $do(.)$ operator

- Pearl introduces causal states using the semantics of an ideal experimental intervention and what he labels the $do(.)$ operator.
- For a graph that includes a casual effect $D \rightarrow Y$, and the directed edge indicates that D is an input in the function, $f_Y(.)$.
- The two probability distributions that define causal effects are $Pr[Y|do(D=1)]$ and $Pr[Y|do(D=0)]$, not $Pr[Y|D=1]$ and $Pr[Y|D=0]$.
- In particular, the average causal effect is $E[Y|do(D=1)] - E[Y|do(D=0)]$, under the assumption that the individual-level causal effect is defined by the individual-level difference induced by the hypothetical intervention, $[y_i|do(d_i=1)] - [y_i|do(d_i=0)]$).

Pre-intervention and under intervention

- Under this setup, observable associational quantities, based on $Pr[Y|D]$, do not necessarily equal causal quantities, based on $Pr[Y|do(D)]$.
- Most importantly, for the basic confounding graph Figure 3.4(a), the associational difference $E[Y|D=1]-E[Y|D=0]$ does not equal the average causal effect defined by the atomic intervention on D, $E[Y|do(D=1)]-E[Y|do(D=0)]$. The confounding from C generates additional dependence between D and Y that enters into the average associational difference, $E[Y|D=1]-E[Y|D=0]$, but not the average causal difference, $E[Y|do(D=1)]-E[Y|do(D=0)]$.
- The reason that C does not enter the under-intervention difference is that D is determined by a hypothetical ideal experiment in this regime. In contrast, in the pre-intervention regime, D is determined by the structural equation $D = f_D(C, e_D)$ that has C as one of its inputs.

The connection to potential outcomes

- The $do(.)$ operator is the exact analog to the superscripts given to potential outcomes in order to designate the underlying causal states that define them.
- Pearl states that the $do(.)$ operator "is a mathematical device that helps us specify explicitly and formally what is held constant, and what is free to vary" (Pearl 2009:358).
- The semantics that accompany the $do(.)$ operator – "ideal experiment" and "atomic intervention" – are Pearl's chosen way to express the idea that all units in the population could be assigned to the causal states in which they are observed (the "factual" ones) or to the causal states in which they are not observed (the "counterfactual" ones) and that causal effects are defined by differences attributable to movement between these alternative states.

How can we ignore the $do(.)$ operator?

- As long as one maintains that it is the $do(.)$ operator that defines causal effects, not associational contrasts such as $E[Y|D=1]-E[Y|D=0]$, the $do(.)$ operator does not need to be represented in the causal graph in any explicit way.
- The primary purpose of the graph is to encode the full set of causal relationships that one assumes characterize a causal effect of interest, so that those causal relationships can be considered in both the pre-intervention regime and the under-intervention regime.
- The $do(.)$ operator is therefore a piece of the underlying structure of the graphical approach, such as the error terms of nonparametric structural equations, which can be brought into the foreground when necessary to explain an identification result.

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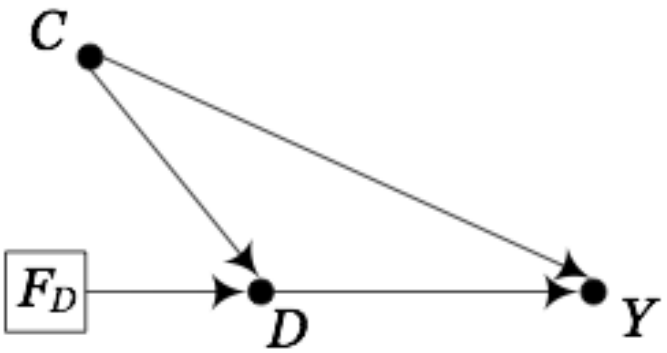
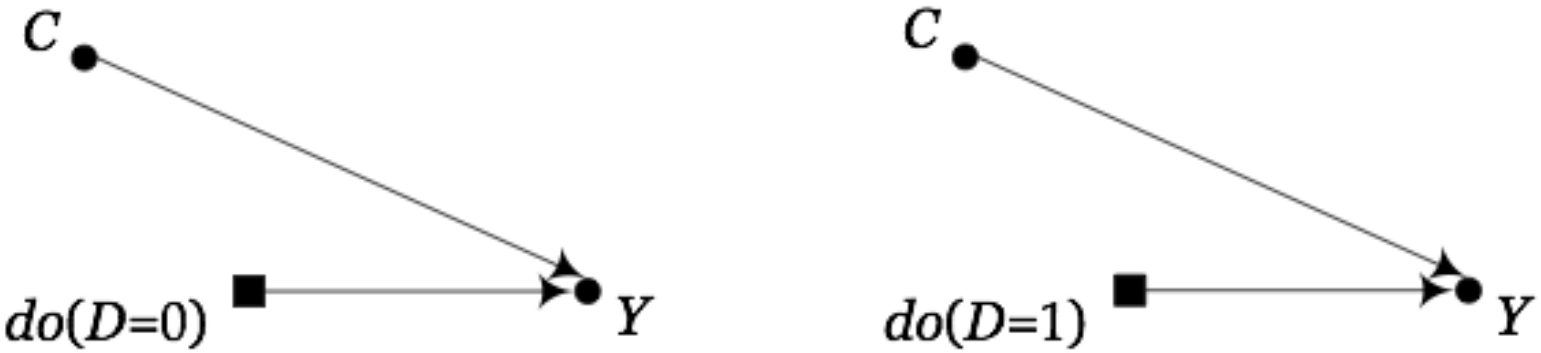
Graphs that represent interventions explicitly



(a) an augmented causal graph with a "forcing" variable that represents an intervention



(b) "mutilated" graphs that demonstrate the *do*(.) operator for the two values of *D*

Criteria for directed graphs that can be anointed causal graphs

A directed graph can be considered a causal graph by Pearl's definitions if

- All variables in the graph, $\{V\}$, are observed (other than those variables implicitly included in the error terms, $\{e_V\}$, that are only revealed under magnification).
- All variables in the graph, $\{V\}$, have error terms, $\{e_V\}$, that are independent of all variables in the graph, $\{V\}$, and that are mutually independent of each other.

and ...

Criteria for causal graphs (cont'd)

- It is reasonable to assume that each variable in the graph, V, can be subjected to a hypothetical intervention, $do(V = v)$, that
  1. replaces the pre-intervention probability distribution of V, $Pr(V)$, with a single intervention value v,
  2. removes the directed edges in the graph that terminate at V, and
  3. changes nothing else in the graph (even though the setting of V to v propagates through the probability distributions of the descendants of V via the directed paths that remain in the graph).

The absence of potential outcome variables from causal graphs

- You cannot intervene on a potential outcome! They already encode the intervention.
- Recall the definition offered in Equation (2.2):

$$Y = DY^1 + (1 - D)Y^0. \tag{3.8}$$

- One trivial way to explain why causal effects such as $D \to Y^1$ are nonsensical is to take Equation (3.8) and rewrite Y^1 in individual realizations as

We can draw graphs to show associations

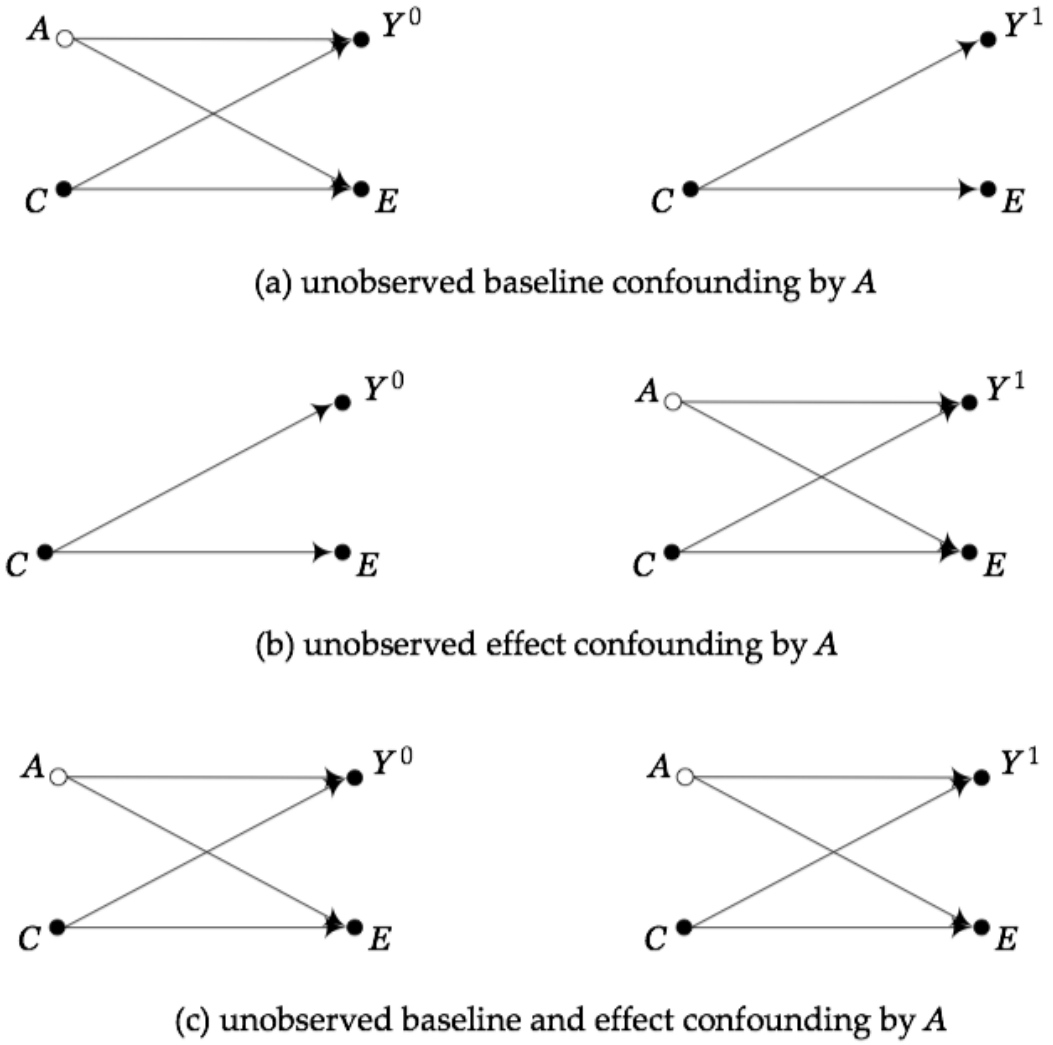


Figure 3.9 Alternative graphs for the joint dependence of a two-valued causal variable for education (E) and potential outcomes for earnings (Y^0 and Y^1) on observed confounders (C) and on an unobserved confounder for ability (A).